Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/29/2010
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Recovery of Musculoskeletal Quantity and Quality upon Multiple Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stefan.judex@sunysb.edu	**Fax:**	FY 631-632-8577
PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	10/01/2008	**End Date:**	09/30/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Clinton (SUNY Stony Brook)		
Grant/Contract No.:	NNX08BA35G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	As space missions become longer, it is apparent that long-term risks of multiple, or even single, exposures to microgravity on crew musculoskeletal health need to be defined and addressed. Even though the magnitude of the long-term deterioration in morphology and strength is largely unknown, it is clear that such risks will be heavily influenced by the genetic make-up of an individual. We have recently identified and characterized a genetically diverse mouse population that, similar to astronauts, includes a diverse range of responses to hindlimb unloading and reambulation. Using this population, here, we will address the consequences of multiple exposures to simulated weightlessness on long-term quantity and quality of trabecular and cortical bone as well as muscle, and investigate how recovery periods between periods of weightlessness can be optimized. Changes in tissue morphology, density, and mechanical properties will be assessed longitudinally in each genetically distinct mouse by in vivo microCT and		

finite-element modeling, allowing for the identification of skeletal baseline indices that modulate changes in tissue morphology and strength during altered levels of loading. Detailed mechanical tests will be performed at the endpoints to assess the long-term quality of the tissue. Together, the proposed studies provide a unique opportunity to establish the range of responses of genetically diverse individuals to multiple unloading and reambulation phases. Optimizing the length and distribution of recovery periods between disuse exposures may ultimately help to maintain long-term musculoskeletal morphology and strength.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Extended periods of unloading are not confined to space flight but also occur during extended bedrest or paralysis on Earth. Therefore, data from our experiments may help understanding the basis by which the musculoskeleton responds to repeated unloading events and eventually lead to more efficacious prophylactics and treatments.

PILOT STUDY
In the proposed experiments (SA 1-2), all disuse mice are subject to the first hindlimb unloading (HLU) cycle at 16wk of age, irrespective of the HLU and reambulation (RA) pattern to which they are assigned. However, the age at which they experience additional HLU cycles varies between different experimental groups. Hence, differences in musculoskeletal observed between mice subjected to multiple disuse periods may be attributed not only to differences in their HLU and RA pattern, but to differences in the age at which the subsequent HLU cycles are applied. A pilot study to elucidate the effects of aging on disused-induced changes in musculoskeletal quantity and quality has now been completed.

Methods

Forty-four adult (16wk) male C75BL/6 mice were assigned to age-matched control and disuse groups subject to single (1x-HLU), double (2x-HLU), or triple (3x-HLU) (n=11/group) hindlimb unloading cycles. During each cycle, mice underwent tail suspension continuously for 2wk. Each exposure to HLU was followed by 4wk of reambulation (RA). 3x-HLU mice were subjected to 2wk of HLU at 16wk, 22wk, and 28wk of age. To disassociate the concomitant effects of advanced age and prior exposures to HLU on the 22wk and 28wk HLU cycles for in the 3x-HLU animals, 2x-HLU mice were exposed to HLU at 22wk and 28wk, while 1x-HLU animals were suspended only at 28wk. All animals were sacrificed at 34wk.

Results

In-vivo microCT – Bone and Muscle: At baseline, trabecular and cortical bone morphology was comparable for all groups. During the course of the 18wk experiment, all mice experienced significant age-related loss in bone volume and BV/TV. This deterioration was compounded by HLU - the rate of bone loss accelerated during unloading periods, with suspended animals losing more bone than both their ambulatory counterparts.

Bone atrophy persisted even during RA periods, though the relative loss during RA was lower than that observed during HLU cycles. Trabecular microarchitecture, including Tb.N, Tb.Sp, and Conn.D also continued to deteriorate throughout HLU and RA periods for all groups. At the time of sacrifice, the differences in these trabecular indices between disuse and ambulatory mice were insignificant. However, the additional erosion of trabecular bone in multiply suspended animals was attributable largely to differences in Tb.Th, which was highly sensitive to weightbearing. Tb.Th decreased significantly during each HLU in all three disuse groups, inhibiting age-related thickening of the trabeculae. Though a significant recovery in thickness was observed during RA periods, the recovery was incomplete, and after multiple unloading and RA cycles, Tb.Th was significantly reduced in 2x-HLU and 3x-HLU animals in comparison to age-matched counterparts. These findings suggest that the trabecular bone remains susceptible to the loss of weightbearing across multiple exposures to disuse.

Like Tb.Th, both Ct.Ar and Ct.Th in the metaphysis were also sensitive to weightbearing, as was the polar moment of inertia in the diaphysis. Unlike the trabecular compartment, in which bone loss continued through RA periods, these indices decreased only during HLU cycles, recovering to or surpassing pre-HLU values after each 4wk RA period. A significant increase in Ct.Ar and Ct.Th from 16wk to 34wk was noted only for ambulatory controls and 1x-HLU animals, indicating that the additional exposures to HLU depressed short-term bone formation within the cortex. This decrease was attributed to an increased erosion of the periosteal surface in the metaphysis. On the other hand, HLU resulted in a net loss of cortical bone in the diaphysis, resulting from an inhibition of periosteal expansion in that compartment.

For trabecular and cortical indices that were responsive to weight-bearing pattern, the magnitude of the response to HLU and RA decreased with the number of prior exposures to HLU and RA, respectively. Consequently, the bone quantity and quality of 2x-HLU and 3x-HLU mice was comparable at sacrifice. On the other hand, after accounting for age-related changes, the age of initial suspension did not significantly affect the magnitude of the response for the initial suspension for nearly all variables, irrespective of whether or not they were responsive to loading. Hence, multiple exposures to disuse significantly erode trabecular and cortical bone quality and quantity. Increasing the number of unloading cycles further deteriorates bone morphology, though the sensitivity to unloading decreases with subsequent exposures to disuse.

Muscle volume was comparable for ambulatory and experimental disuse groups at baseline. Though both control and HLU animals experienced muscle loss in the first few weeks of the study, disuse groups become more sensitive to changes in weight-bearing after that. However, the total change in muscle volume from baseline to sacrifice within groups was comparable, and at sacrifice, the differences between groups were comparable.

Histomorphometry and Histology

Indices of bone formation were sensitive to disuse – MS/BS, MAR, and BFR/BS decreased with increasing HLU cycles with significant differences in these parameters between control and 3x-HLU groups. The percent of trabecular, endosteal or periosteal surfaces at which osteoclastic activity was evident (Oc.S/BS) was not significantly different between ambulatory and disuse groups in either the metaphysis or the diaphysis, though there was a trend towards increasing osteoclastic activity with increased # exposures to disuse in the trabecular compartment.

In-vivo microCT – Abdominal Tissue: At 28wk, experimental and control mice had comparable body mass, lean mass,

Task Progress:

and aVMD. However, TAT, SAT, and VAT were significantly lower in the 2x and 3x-HLU mice, both of which had been previously suspended, suggesting that prior exposures to HLU had suppressed fat accumulation in disuse mice. The fat composition was comparable - prior to HLU at 28wk, VAT was approximately 70% of TAT for all groups. During the HLU between 28wk and 30wk, disuse mice experienced weight loss, which was accompanied by a greater concomitant % loss of TAT, SAT, VAT, lean mass and vertebral density after HLU. The decrease in absolute VAT volume was also accompanied by a decrease in %VAT to 60% post-HLU. Further, the % lean mass in the abdomen increased in disuse mice.

Since the magnitude of the response to HLU decreases with increasing exposures to disuse, the changes in adiposity and vertebral density immediately following HLU were greatest in 1x-HLU mice, though all disuse mice experienced a significant change in these indices relative to baseline, irrespective of whether they had 0, 1, or 2 prior exposures to disuse. Further, changes in vertebral bone density and all three fat compartments in response to HLU were significantly correlated ($R2=0.7406$, $p < 0.001$). Though disuse HLU mice largely recovered TAT, VAT, and SAT after 4wk of RA, leading to values comparable to age-matched animals, only 2x- and 3x-HLU groups achieved recovery of BMD by 34wk. Regardless of the extent of recovery, the ratio of VAT to TAT returned to 70% for all groups.

In contrast to clinical studies of bed rest or spinal injury, in which abdominal and marrow fat increase, a decrease in adiposity was observed during HLU in our study. This may be attributed in part to lower food intake during HLU, which preferentially reduced visceral fat and thereby altered the composition of abdominal adiposity. In contrast to indices of femoral bone morphology, all abdominal indices returned to values comparable to age-matched control for 2x- and 3x-HLU mice, but TAT did not fully recover post-RA in 1x-HLU mice. The results suggest that only the initial cycle of HLU reduces adiposity permanently, with complete recovery after multiple exposures that match the recovery of vertebral bone.

SPECIFIC AIM 1

Methods: 12wk old male CH3 and female BALB mice were used to generate an F1 (C3H x BALB) population which was then mated to breed the second generation F2 cross. One hundred twenty adult (16wk) F2 mice were divided into four groups (n=15 male/group, n=15 female/group). Three experimental groups were exposed to a total of 4wk of HLU and 12 wk of RA. The 1 x 4wk group was exposed to a single, continuous 28d HLU cycle (at 16wk) followed by a single 12wk RA cycle, the 2 x 2wk group was subject to two 14d HLU cycles (16wk and 24wk), with each HLU period followed by 6wk of RA, and the 4 x 1wk group was exposed to four 7d HLU cycles, with each HLU cycle followed by 3wk of RA. All mice were sacrificed at 32wk.

Results

In-vivo microCT – Bone: Both the male and female F2 mice population exhibited heterogeneity in baseline skeletal bone morphology. Males and females from all groups, including age-matched controls, experienced a significant loss of metaphyseal trabecular bone over the 16wk experimental period, with a concomitant decrease in Tb.N and Conn.D, and an increase in Tb.Sp. A greater decrease in BV/TV was observed for disuse groups of the F2 males, attributed primarily to a greater loss in Tb.N. Similar to the pilot study, bone loss continued during the RA periods. In mice that were suspended, Tb.Th was the only morphologic index that reversed upon HLU, increasing during RA cycles. Relative to age-matched controls however, this recovery was incomplete in all of the 3 experimental groups, leading to a post-RA bone morphology that was distinct from that observed prior to HLU in both genders. Between the 3 disuse groups however, there no significant differences in morphometric indices.

Diaphyseal Ct.Ar was also mechanosensitive. However, only female F2 mice experienced a complete recovery of disuse-induced atrophy during RA periods. Males subject to HLU exhibited much lower cortical mass after 16wk compared to controls, attributed to an enhanced expansion of the marrow cavity and a reduced expansion of the periosteal envelope.

Muscle mass was also highly sensitive to altered loading patterns. Disuse also inhibited an age-related accrual of muscle mass, and despite a modest recovery during RA periods, the total increase in muscle mass in all disuse groups was significantly lower than that observed in their age-matched counterparts . For most morphological indices, the magnitude of the response to unloading decreased with subsequent exposures in multiple-unloaded animals.

These data show that the musculoskeleton of the F2 mouse population was highly sensitive to disuse, irrespective of whether the unloading period was partitioned into 1, 2, or 4 cycles. The sequence of disuse followed by reambulation resulted in trabecular architectures that were markedly distinct from baseline and age-matched controls, even when the overall quantity of the bone was comparable between the groups. Interestingly, there were no interactions between gender and un- and reloading, even though altered hormone levels are known to contribute to gender-specific differences during aging and disuse. Further, the differences in bone morphology between groups subject to 1, 2, or 4 cycles of HLU were insignificant. Together, these results suggest that the extent of femoral bone-loss is primarily modulated by the total duration, rather than the schedule of un- and reloading.

In-vivo microCT – Abdominal Tissue: Both male and female F2 mice had a heterogeneous distribution in body mass at baseline, but the average body mass between groups was comparable. 1x-4wk and 2x-2wk mice experienced an accrual of body mass that was comparable to that observed in control mice. However, frequent exposure to HLU suppressed this age-related increase in body mass in 4x-1wk mice. Analogous changes were noted in several abdominal fat tissues. Though the % increase was distinct between genders, control, 1x-4wk, and 2x-2wk mice all underwent a comparable increase in total, visceral, and subcutaneous fat through the course of the 16-wk experiment. In 4x-1wk females, recurring HLU inhibited this accrual of abdominal fat tissue, while in 4x-1wk males disuse actively reduced of abdominal fat between baseline and sacrifice.

During the 16wk experimental period, the composition of the abdominal tissues also changed – control, 1x-4wk and 2x-2wk females all experienced an increase in the ratio of VAT to TAT, accompanied by a decrease in the percent of lean mass within the abdomen. However, in 4x-1wk females, the abdominal fat tissue composition was comparable at baseline and sacrifice, with an increase in percent lean tissue. On the other hand, male F2 mice underwent an insignificant change in fat tissue composition between baseline and sacrifice, irrespective of whether they were ambulatory or subject to disuse. While control, 1x-4wk, and 2x-2wk males experienced a small but significant decline in percent lean tissue mass, like their female cohorts, percent lean tissue mass increased in 4x-1wk males. Further, aVMD for 4x-1wk males was also significantly lower at sacrifice, and the decrement was significantly less than that observed in control, 1x-4wk, and 2x-2wk cohorts.

	In contrast to the changes observed in the appendicular skeleton, these findings suggest that systemic effects of disuse are influenced by primarily modulated by frequency, rather than the duration, of disuse. In the 4x-1wk group, the repetitive exposures to disuse may disrupt cellular activity and metabolism sufficiently that, unlike 1x-4wk and 2x-2wk animals, which have 12wk and 6wk RA periods, the shorter 3wk RA time in the 4x-1wk mice may not permit the recovery of these parameters back to levels observed in age-matched controls before exposure to the subsequent HLU cycle. SPECIFIC AIM 2 Ninety female F2 mice have been bred for the experiment proposed in SA2. The baseline activity of the 3x-3wk group has been recorded with the activity monitoring cages, and this group will begin its first HLU starting August 2, 2010. The study is scheduled to continue through July, 2011.
Bibliography Type:	Description: (Last Updated: 01/16/2018)
Abstracts for Journals and Proceedings	Gupta S, Vijayaraghavan S, Tuthill A, Judex S. "Recovery of Abdominal Fat and Vertebral Bone Density after Multiple Exposures to Mechanical Unloading." To be presented at the Annual American Society of Bone and Mineral Research Meeting, Toronto, Ontario, Canada, October 5-19, 2010. Proceedings of the Annual American Society of Bone and Mineral Research Meeting, Toronto, Ontario, Canada, 2010. , Oct-2010
Abstracts for Journals and Proceedings	Gupta S, Vijayaraghavan S, Tan L, Judex S. "Disuse-Induced Changes in Cortical Bone Morphology are Independent of Gender in a Genetically Diverse F2 Mouse Population." 56th Annual Meeting of the Orthopaedic Research Society, March 6-9, 2010, New Orleans, LA. Proceedings of the 56th Annual Meeting of the Orthopaedic Research Society, March 6-9, 2010, New Orleans, LA. , Mar-2010
Abstracts for Journals and Proceedings	Gupta S, Uzer G, Judex S. "Bone Atrophy and Recovery upon Multiple Exposures to Mechanical Unloading." 31st Annual Meeting of the American Society of Bone and Mineral Research, Denver, CO, September 2009. Proceedings of the 31st Annual Meeting of the American Society of Bone and Mineral Research, Denver, CO, September 2009. , Sep-2009